

Aggregate Class Cheat Sheet

m `Aggregate(name, exp_el, exp_premium, exp_lr, exp_en, exp_attachment, exp_limit, exp_years, exp_rate, sev_name, sev_a, sev_b, sev_mean, sev_cv, sev_loc, sev_scale, sev_xs, sev_ps, sev_wt, sev_lb, sev_ub, sev_conditional, sev_signed, sev_reflect, sev_pick_attachments, sev_pick_losses, occ_reins, occ_kind, freq_name, freq_a, freq_b, freq_zm, freq_p0, freq_pin_mean, wait_*[0], agg_reins, agg_kind, reins_bucket, dsev_bucket, value_type, approximate, label, label_map, note, hints, tags, doc)`

The constructor follows the DecL clauses (exposure, severity, occurrence reinsurance, frequency, aggregate reinsurance, approximate). Usually built via `build('agg ...')`. The following tables show all **m** methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

`name` (identity), `label` (display), `limit`, `attachment`, `frequency`, `sevs` (list of Severity), `en` (expected claims), `n` (frequency), `base_mean` (un-modified $E[N]$, before `zt/zm`), `value_type` (loss / payoff), `exp_premium`, `exp_lr`, `exp_rate`, `exp_years`, `sev_pick_attachments`, `sev_pick_losses`, `program`, `pprogram` (canonical DecL), `spec`, `spec_ex`
Derived objects: **m** `make_pnl` (wrap as the risky leg of a PnL), **m** `create_frequency` (claim count as its own Aggregate)

2. Update

`log2`, `bs` (bucket size), `sev_calc` (round / forward / backward), `discretization_calc` (distribution / survival / both), `normalize`, `padding`, `dsev_bucket`, `reins_bucket` (atom placement schemes), `x_min`, `x_max`, `i0` (window), **m** `update`, **m** `update_work`, **m** `discretize`, **m** `picks` (hit layer loss picks), **m** `rescale` (homog. sev / inhomog. freq), **m** `snap`, **m** `unwrap` (undo FFT wrapping)

3. Moments

actual = analytic, est = estimated from the FFT grid
`actual_m`, `actual_cv`, `actual_sd`, `actual_var`, `actual_skew`, `est_m`, `est_cv`, `est_sd`, `est_var`, `est_skew`, `sev_m`, `sev_cv`, `sev_sd`, `sev_var`, `sev_skew`, `est_sev_m`, `est_sev_cv`, `est_sev_sd`, `est_sev_var`, `est_sev_skew`, **m** `freq_pmf`

4. Statistical functions

m `cdf`, **m** `sf`, **m** `pdf`, **m** `pmf`, **m** `q` (lower quantile = VaR), **m** `tvar`, `sev` (exact severity cdf / sf / pdf), **m** `q_sev`, **m** `tvar_sev`, **m** `var_dict`^[1], `prob_eq_0` ($\Pr(X=0)$), **m** `sample`

5. Validation

`valid`, `validation_eps`, `validation_df` (moment vs. estimate, the QA view), `validation_explanation`, **m** `aggregate_error_analysis` (agg error over bs), **m** `severity_error_analysis` (trunc. / discr. by component), `bs_window_df`, `bs_description`, `bs_explanation`, `tail_class` ((freq, sev, agg) triple), `tail_behavior_df`, `tail_description`, `tail_explanation`, `bounded`

6. Output dataframes

`density_df`^[1] (main output), `density` (live rows only), `summary_df` (at a glance: moments + percentiles), `stats_df`, `info`, `tail_df`, **m** `tail_periods_df` (return periods), **m** `focus` (central window holding $1 - p$ of the mass), `agg_density`, `sev_density`, `sev_density_df`, `ftagg_density` (FT), `xs`, `xs_sev`, `figure` *see also Reinsurance*

7. Reinsurance

`occ_kind`, `occ_reins`, `agg_kind`, `agg_reins`, `agg_density_gross/ceded/net`, `sev_density_gross/ceded/net`, `occ_ceder`, `occ_netter`, `agg_ceder`, `agg_netter`, **m** `apply_occ_reins`, **m** `apply_agg_reins`, `reins_kinds`, `reins_description`, `reins_summary_df` (per stage, the daily driver), `reins_density_df`, `reins_stats_df`, **m** `occ_bivariate` (joint law of two of gcns; see the BivariateAggregate card)

8. Visualization

m `plot`, **m** `reins_occ_plot`, `figure`

9. Risk and pricing

m `price`, **m** `price_ccoc`, **m** `price_pentagon`, **m** `price_pentagon_ex` (complete the pricing octet), **m** `prob_loss_assets` (consistent (p, L, a) triple), **m** `apply_distortion`, **m** `calibrate_distortions` (standard set to a CoC target), **m** `pollaczek_khinchine` (probability of eventual ruin, Poisson), **m** `wiener_hopf` (ruin for a renewal / Sparre-Andersen model)

10. Approximations

`approximation` (the stored kind: exact / sgamma / slognorm), **m** `approximate` (fit by moment matching)^[2], **m** `entropy_fit`, `fzapprox`

11. Meta

m `help`, **m** `json` (persist), **m** `format_program`, `pprogram_html`, `note`, `tags`, `doc`, `hints` (the DecL trailer), `label_map`, `labels`, `renamer`, `use_labels` (the label surface)

Notes:

[0]: `wait_*` is the renewal (Sparre-Andersen) waiting-time family: `wait_name`, `_a`, `_b`, `_mean`, `_cv`, `_loc`, `_scale`, `_xs`, `_ps`, `_wt`, `_lb`, `_ub`, `_conditional`, `_attachment`, `_limit`. It pairs with `exp_years`, never with a frequency clause.

[1]: matches Portfolio. [2]: `approximate` is the *action*; `approximation` is the *stored kind* set by the DecL clause.

Abbreviations: gcns = gross (subject), ceded, net; actual = analytic, est = FFT estimate.